AA

SAVITRIBAI PHULE PUNE UNIVERSITY

PCC-201-COM-DS

Sem:3 7022416

Seat No S500220261
Stk. No 3497263
Sub. DS
Centre 4022

Examination	SC (2024) Comp Engg
Day & Date	monday - 15/12/24
Subject	Data Structure
Paper No.	[6582]- 236 Sec
Medium of Answer	English

Seat No. : In figure & In words

	5	0	0	2	2	0	2	6

S Five zero zero two two zero Two six one

Signature of Candidate	Pooja
Signature of Supervisor	15/12/24

(in Red Ink)

Total	Marks in Figure	Marks in Words	Sign
Examiner	42	Four Two	
Moderator			

Instruction to Candidate

1. Candidate has to confirm seat number, subject and centre number printed on Bar code and write it on attendance sheet.
   विद्यार्थ्याने प्रथम बार कोडवरील आसन क्रमांक, विषय व केंद्र क्रमांक तपासून योग्य असल्याची खात्री करावी आणि उपस्थिती पत्रकावर नोंदवावी.
2. Paste Bar Code in prescribed space.
   उत्तरपत्रिकेवरील विहित जागेतच बार कोड लावावा.
3. Do not write anything on Bar code sticker, otherwise it will be treated as unfairmeans.
   बार कोड स्टिकरवर काहीही लिहू नये, अन्यथा परीक्षा गैरप्रकार समजला जाईल.

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12		
Total in Figure	42	
Total in Words	Four Two	
Signature		

१. विद्यार्थ्याने उत्तरपत्रिकेच्या मुखपृष्ठावर तसेच उपस्थिती पत्रकावर विहित जागेत आसन क्रमांक अंकात व अक्षरात बिनचूक लिहून स्वाक्षरी करावी.

२. उत्तरपत्रिकेवर फक्त निळ्या अथवा काळ्या शाईचा उपयोग करावा, अन्यथा उत्तरपत्रिकेचे मूल्यमापन केले जाणार नाही.

३. उत्तरपत्रिकेच्या पृष्ठक्रमांक ३ पासून लिहिण्यास प्रारंभ करावा.

४. संबंधित प्रश्नाचे अथवा उपप्रश्नाचे उत्तर जेथून सुरू होते तेथेच समासात प्रश्न क्रमांक, उपप्रश्न क्रमांक अचूक व स्पष्ट लिहावा, यासाठी वेगळ्या शाईचा उपयोग करू नये.

५. प्रत्येक पानाच्या दोन्ही बाजूस लिहावे, उत्तरपत्रिका किंवा पुरवणी उत्तरपत्रिकेचे कोणतेही पान फाडू नये, फाडल्यास परीक्षा गैरप्रकार समजून पुढील कार्यवाही करण्यात येईल.

६. पेपर संपण्यापूर्वी १० मिनिटे अगोदर इशारा घंटा होईल, त्यानंतर विद्यार्थ्याने उत्तरपत्रिका व पुरवणी उत्तरपत्रिकेवर होलोक्राफ्ट स्टिकर विहित जागेवरच लावावा.

७. कॉपी करणे किंवा दुसऱ्याच्या नावावर परीक्षेस बसणे यांसारख्या कृती 'महाराष्ट्र-प्रीव्हेन्शन ऑफ मालप्रॅक्टिस ॲक्ट युनिव्हर्सिटी, बोर्ड ॲण्ड अदर स्पेसिफाईड एक्झामिनेशन्स ॲक्ट, १९८२' (सा.फु.पु.वि. चा अध्यादेश क्रमांक ९) त्यानुसार संमत केलेला कायदा या अन्वये दंडही असेल.

Candidate shall fill all information about seat number, paper etc. in prescribed space and sign on the answer book and attendance sheet.

Candidate shall use blue or black ink only. Otherwise answer book will not be evaluated.

Candidate shall start writing answer from page no. 3 of the answer book.

Candidate shall mention question number, sub question number correctly at the beginning of the same and shall not use ink other than blue or black.

Candidate shall write on both sides of pages and shall not tear off any page, it will be treated as unfairmeans.

Warning bell will be given before 10 minutes of the concluding time. Candidate shall paste Holocraft Sticker at appropriate space on the answer book.

An Act of Copying or Impersonations at an Examination is Punishable under 'The Maharashtra Prevention of Malpractices at University, Board and Other Specified Examinations Act, 1982' (Ordinance 9 of SPPU). The Act passed to the effect.

Examiner and Moderator has to write marks on all given appropriate place only. Examiner should give assessment tick(✓) or (×) in the margin.

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1				
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Total	42			

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Q2.

(a) Comparison of static and dynamic data structure:

Static	Dynamic
• A type of data structure having fixed size.	• A type of data structure which doesnot have fixed size.
• A fixed memory is allocated at the time of initiation.	• Memory is allocated dynamically or real time.
• Eg: Array	• Eg: Linked List.
• Easy to implement	• It is not easy to implement.

Significance of persistent and Ephemeral data structure:

Persistent :

- A type of data structure which can withstand their original state.

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- For eg: Stack.

8
9
10

It is a Stack having elements 10, 9 & 8.

Now if we want to pop one element lets say, 8

8
9
10

9
10

Now also the remaining sequence is same.

Ephemeral data structure

- A type of data structure which cannot withstand their original state after doing some operations.

- For eg: Queue.

- | 10 | 9 | 8 | It is a queue having elements 10, 9 & 8.

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- Now if we dequeue one element then,

| 9 | 8 |

It loses its original state.

- This is the significance of ~~eph~~ ephemeral data structure.

(b) Step count method:

Code	frequency
n = 100	1
for num in range(1,n):	n
if num%2 == 0:	n·n
print (num)	1
else:	1
print (num)	1

Total frequency ⇒ Total ~~no.~~ complexity of steps

$= 1+n+n\cdot n+1+1+1$

$= n^2+n+4$

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- The above algorithm possess quadratic time complexity because the highest power of complexity obtained from step count method is (2).

- Above algorithm is a linear algorithm because it uses linear data structure. There is no non linear type of data structure used above.

* Various algorithmic strategies used in data structure are as follows:

(i) Greedy algorithm

(ii) Backtracking

(iii) Divide and conquer.

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Divide and conquer:

- It is a algorithmic strategy which usually divide the problem into smaller subproblems and then solve them.

- After solving each block of divided problem, we recombine them.

- Steps involved in divide and conquer technique are:

(i) Divide : Divide the problem.

(ii) Conquer : solve the subproblems

(iii) Recombine : Combine all solved pieces of problem.

- Eg: merge sort

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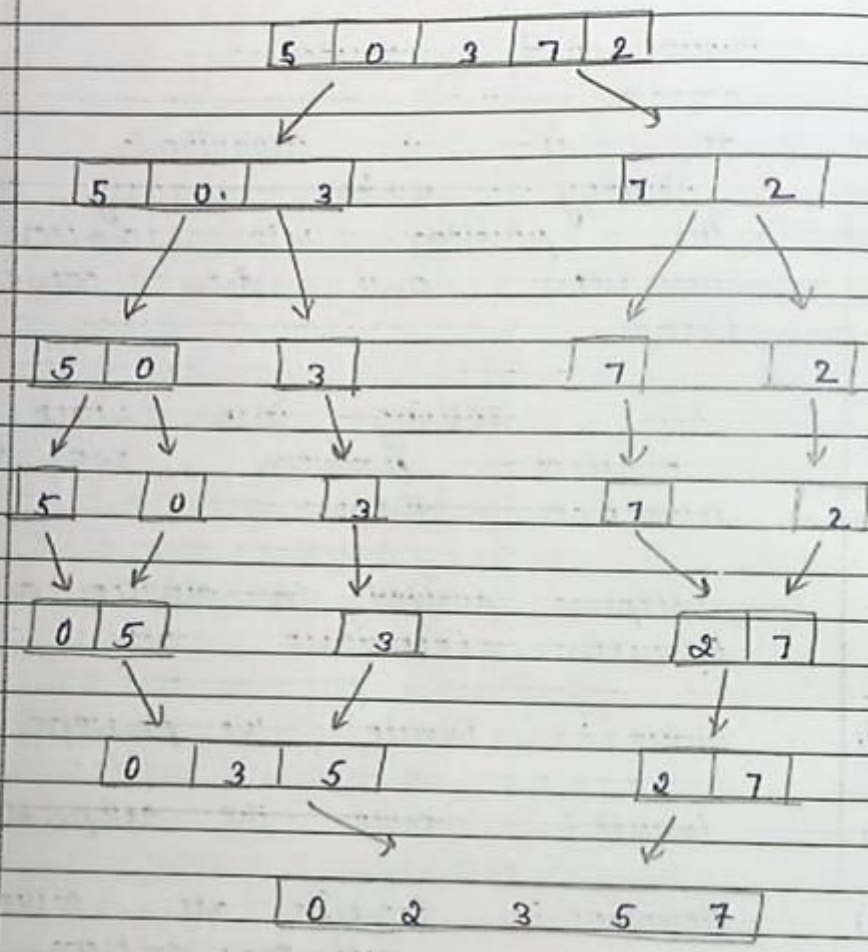

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Q3.

(a) Define terms:

(i) Space matrix:

- A sparse matrix is a ~~matrix~~ data structure which uses array for initialisiation and creation.
- In python matrix is 2-D array.
- It is used to store data which is in form of rows and columns.
- a[i][j] = arr[4][5]

(ii) Sorting stability:

- Sorting stability is a stability which retain the sequence of two similar elements even after operations like sorting.
- For eg:

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Unsorted array:

5 3 3 0 9

After sorting:

0 3 3 5 9

- Here before sorting and after sorting the position of 3 sequence in list remains same.
- This is stability.

(iii) Internal and external sort.

Internal sorting ~~External sort~~

- A type of sorting which uses main memory for its sorting purpose.

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- Internal sorting uses RAM for storage.
- For eg: Bubble sort, Insertion sort, etc.

External sorting

- A type of sorting which due to lack of main memory, uses secondary memory for sorting is called external sorting.
- Very large data can be sorted by external sorting.
- For eg: merge sort.

(b) Binary search algorithm:

9, 17, 23, 28, 45, 50, 57, 76, 90, 1000

- Binary search algorithm is only applicable for sorted array.

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Algorithm:

- Divide the array in half.
- check if mid element is key.
- If not, then check:

  if key < mid :
  high = mid-1

  else :
  low = mid+1

  where, low = 0
  high = ~~dat~~ len(arr)
  mid = (high+low) / 2.

- As given question, keys are 23 and 90.

(i) key = 23.

9 17 23 38 45 50 57 76
5 6

90 100

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pass 1: mid = 50

23 < 50

Then we will conside first half,
high = 45

9 17 23 38 45

Pass 2: mid element = 23

23 = 23

match found in 2nd pass.

(ii) key = 90

pass 1 : mid = 50

50 < 90

considering second half of array,
low = 57

57 76 90 100

Pass 2: mid element = 90

90 = 90

match found in 2nd pass.

Time complexity : O(logn)

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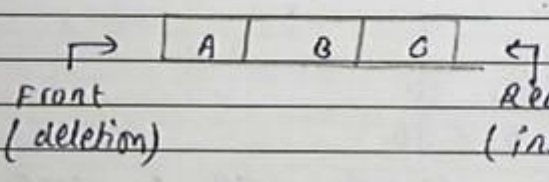
Q6.

(a) Define and draw:

(i) Queue

- Queue is a type of data structure which uses principle of FIFO (First in First out).

- It have two ends that is, enqueue from rear dequeue from front.



(ii) Circular Queue:

- Circular queue is a type of queue in which last element is ~~connect~~ connected to first element.

- It helps to reduce memory wastage.

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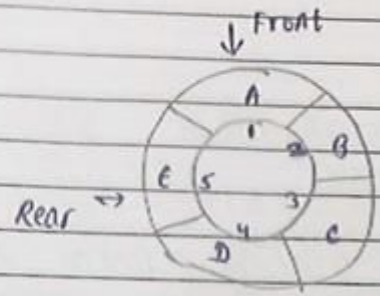

Circular queue.

(iii) Dequeue

- Dequeue is a major operation of queue used to delete elements.

- Dequeue is done from front end of queue.

| 10 | 0 | 3 |

operation : dequeue executed

| 0 | 3 |

10 is deleted because it was front element.

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(E) Stack implementation using linked list: pseudocode

(i) Create : creating linked list to perform operations

Pseudocode:

- ~~def __init__(self):~~
- ~~self.data = data~~
- ~~def __init__(self,head):~~

- def __init__(self):
- self.data = data
- self.next = None
- def __init__(self,head):
- self.head = None

(ii) Push : Insert data elements in linked list

- def insert(head, x):
- new = node(x)
- ~~if head is None:~~
- ~~return head~~
- ~~temp = head~~
- value stored as new.

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(iii) Pop : delete last element.

```
def delete (self, head):
    if head is none
        return none
    temp -> head
    ~~temp=temp.next and~~
    ~~if a temp:~~
        ~~temp=temp.next~~
    while temp.next and temp.next.data != key :
        temp = temp -> next
    Null temp.
```

(iv) Display: Traverse all elements.

```
def display (node):
    temp -> head
    temp -> temp.next
    print ( temp.data)
```

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Q 8.

(a) short note on three hashing techniques:

(i) Division method:

- It is a hashing technique used to put element in hash table by using modulo division

(ii) Chaining

- In this type of hashing, elements are placed at bucket having remainder of division method.

- chaining can be ~~with~~ done with replacement or without replacement.

- Eg: 21, 2, 33, 61, 4

key mod 5

	with	without	
0	4	4	-1
1	21	21	-1
2	2	2	-1
3	33	33	-1
4	61	61	0

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(iii) Linear probing:

- Elements are directly placed according to hash function.
- eg: 21, 2, 33, 61

key used

(collission)		
0	61	61 % 4 = 1
1	21	21 % 4 = 1
2	2	2 % 4 = 2
3	33	33 % 4 = 3

(iv) Quadratic probing:

- In this type of hashing we first find index by normal hash function and then use formula:
  (Hash(key) + i^2) % size.
- It is way of reducing collisions.
- eg: 21, 2, 33, 61, 50

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key mod 5

0	61
1	21
2	2
3	33
4	50

(i) $21 \% 5 \Rightarrow 1$

(ii) $2 \% 5 = 2$

(iii) $33 \% 5 = 3$

(iv) $61 \% 5 \Rightarrow 1$ collision

$i = 0, 1, 2, \ldots$

$(61 + 0^2) \% 5 = 1$

$(61 + 1^2) \% 5 = 2$ collision

$(61 + 2^2) \% 5 = 0$

(v) $50 \% 5 = 0$ collision

$(50 + 0^2) \% 5 = 0$ collision

$(50 + 1) \% 5 = 1$ collision

$(50 + 2^2) \% 5 = 4$

(b) (i) Linear probing:

4371, 1323, 6173, 4199, 4344, 9679, 1989

size = 10

Hash function = key % 10

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(i) 4371 % 10 = 1

0	1	2	3	4	5	6	7	8	9
	4371								

(ii) 1323 % 10 = 3

0	1	2	3	4	5	6	7	8	9
	4371		1323						

(iii) 6173 % 10 = 3

collision occurs → next ~~pos~~ empty slot

0	1	2	3	4	5	6	7	8	9
	4371		1323	6173					

(iv) 4199 % 10 = 9

0	1	2	3	4	5	6	7	8	9
	4371		1323	6173					4199

(v) 4344 % 10 = 4 → collision

0	1	2	3	4	5	6	7	8	9
	4371		1323	6173	4344				4199

(vi) 9679 % 10 = 9 → collision

0	1	2	3	4	5	6	7	8	9
9679	4371		1323	6173	4344				4199

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(iii) 1989 % 10 = 9 → collision

0	1	2	3	4	5	6	7	8	9
9199	~~1989~~ 4371		1323	6173	4344				4199

(iv) Quadratic probing:

(i) 4371 % 10 = 1

(ii) 1323 % 10 = 3

(iii) 6173 % 10 = 3 → collision

Quadratic probing : (Hash(key) + i^2) % size

(i = 0, 1, 2, 3, ...)

so,

$(6173 + 0^2) \% 10 = 3$ ($i=0$)

$(6173 + 1^2) \% 10 = 4$ ($i=1$)

(iv) 4199 % 10 = 9

(v) 4344 % 10 = 4 → collision

so,

$(4344 + 0^2) \% 10 = 4$ ($i=0$)

$(4344 + 1^2) \% 10 = 5$ ($i=1$)

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(vi) 9679 % 10 = 9 → collision
So,

(9679 + 0²) % 10 = 9 (i=0)
(9679 + 1²) % 10 = 0 (i=1)

(vii) 1989 % 10 = 9 → collision
so,

(1989 + 0²) % 10 = 9 (i=0)
(1989 + 1²) % 10 = 0 (i=1)
(1989 + 2²) % 10 = 3 (i=2)
(1989 + 3²) % 10 = 0 (i=3)

Linear probing index	data	Quadratic Probing index	data
0	9679	0	9679
1	4371	1	4371
2	1989	2	-
3	1323	3	1323
4	6173	4	6173
5	4344	5	4344
6	-	6	-
7	-	7	-
8	-	8	1989
9	4199	9	4199

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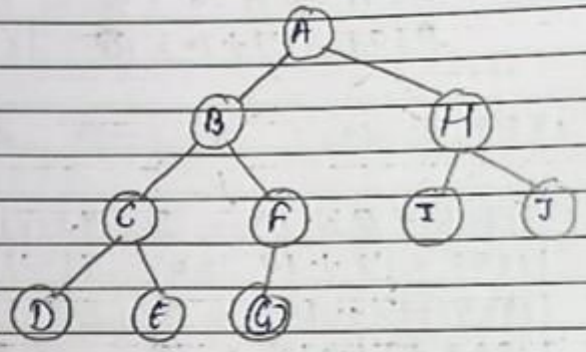
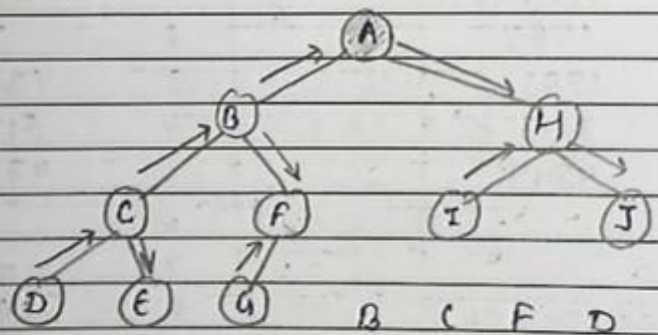
Q10.

(a) Inorder, Preorder and postorder sequence:



(i) Inorder :

Left child → Root → Right child



B C F D E G A
H I J

sequence : ~~D E C B G F A~~
~~H I J~~

~~D E C G F B A~~
~~H I J~~

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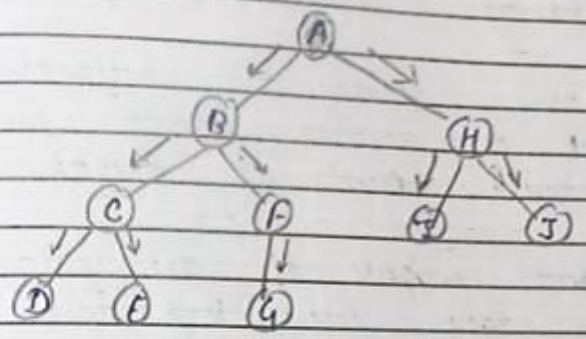
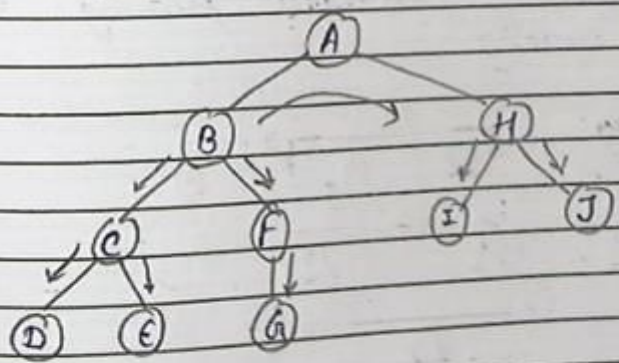
(ii) Preorder :

Root → Left child → Right child



Sequence :- A B C F D E G H I J

(iii) Post Order:
Left child → Right child → Root



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sequence: D C F D E G H I J A

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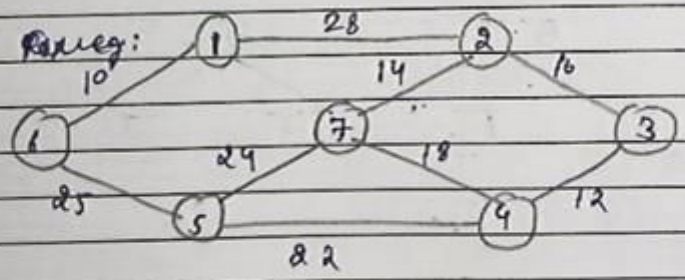
(b) Krustal algo n' thm:

- Start with a vertexes pair having minimum weight
- Now search for adjacent pair of vertices having lowest weight around its neighbours.
- Add weights of all joining the edges till (V-1) edges.
- Cycle should not be formed in between.
- We are putting weights in ascending order and adding without forming a cycle.
- Prog:

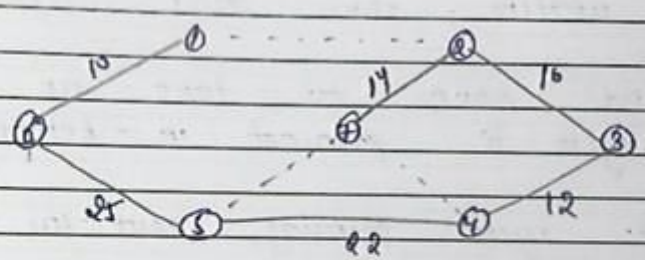


For cost of MST given:

- weights in ascending order:

10 ~~12~~ ~~14~~ ~~16~~ (18) 22 (24) ~~25~~ (28)

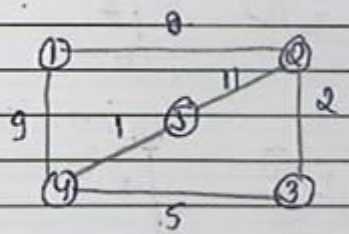
- Now we will connect the vertices:



Total cost : 10 + 12 + 16 + 22 + 25 + 14

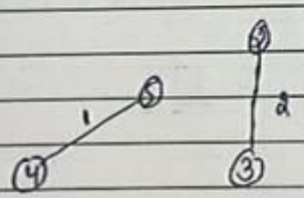
: 99

Stepwise explanation of kruskal by example:

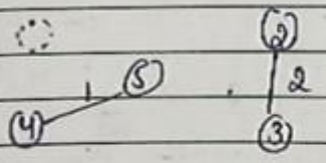


- In this graph first, arrange the weights in ascending order of weights:
- weights : 1 2 5 8 9 11
- Now we'll start connecting vertices from first weight.
- we have to take no cycle is formed in between.
- If cycle forming, leave the weight.

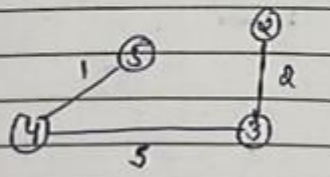
(i) 1 :



(ii) 2 :



(iii) 5 :



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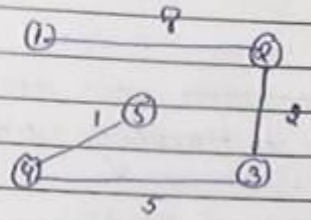
(iv) 8:



(v) 9: skip

(vi) 11: skip

(vii) Total cost of spp spanning tree :
= 1 + 2 + 5 + 8
= 16

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Q6

(b) Stack implementation pseudocode:

(i) Create : Initialising linked list

```
def __init__(self):
    self.data → data
    self.next → none
def __init__(self, head):
    self.head → done
```

(ii) Push : In stack we push data from first front.

```
def insert(head, x):
    If head is none:
        return head
    else:
        head.next → new
        new → node(x)
```

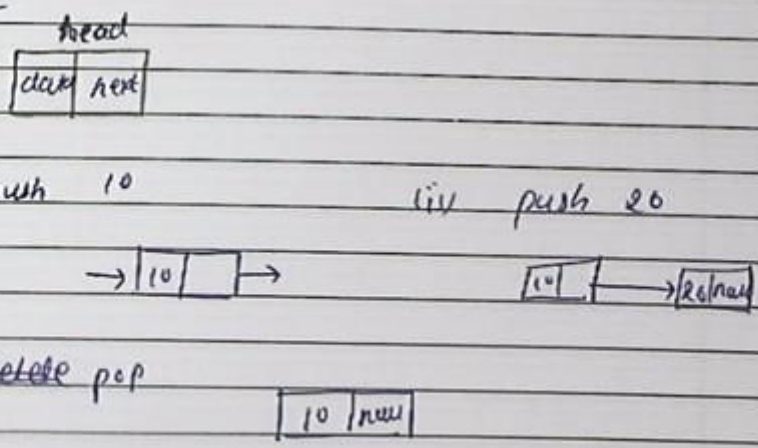
(iii) pop : In stack, element last added was deleted first.

```
def delete( self, head):
        if head is none:
                Return head
    temp → head
while ~~tep~~ temp.next and temp.next.
        data != key
        temp → temp.next
    Null temp.
```

iv) Display: In this we traverse through all elements and print them.

```
def display( node):
    temp → head
    while temp:
        temp → temp.next
    print (temp.data)
```

Eg:-



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	Q.No.						TOTAL
E							
M							

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	Q.No.							TOTAL
E								
M								

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	Q.No.						TOTAL
E							
M							

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प्र. क्र./Q.No.

SPPU-36/36

	Q.No.						TOTAL
E							
M							

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